



Universität Stuttgart

Thesis Title

Von der Fakultät für Informatik, Elektrotechnik und Informationstechnik der
Universität Stuttgart zur Erlangung der Würde eines Doktors der
Naturwissenschaften (Dr. rer. nat.) genehmigte Abhandlung

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Tag der mündlichen Prüfung: unbekannt

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KURZFASSUNG

Kurzfassung der Arbeit.

ABSTRACT

Write an abstract for your work. Replace each of the points below with one sentence (two if you must) and you have your abstract. Write it when you finished your entire report. ¹

Introduction. In one sentence, what's the topic? Phrase it in a way that your reader will understand. If you're writing a PhD thesis, your readers are the examiners – assume they are familiar with the general field of research, so you need to tell them specifically what topic your thesis addresses. Same advice works for scientific papers – the readers are the peer reviewers, and later others in your field interested in your research, so again they know the background work, but want to know specifically what topic your paper covers.

State the problem you tackle. What's the key research question? Again, in one sentence. (Note: For a more general essay, I'd adjust this slightly to state the central question that you want to address) Remember, your first sentence introduced the overall topic, so now you can build on that, and focus on one key question within that topic. If you can't summarize your thesis/paper/essay in one key question, then you don't yet understand what you're trying to write about. Keep working at this step until you have a single, concise (and understandable) question.

¹<https://www.easterbrook.ca/steve/2010/01/how-to-write-a-scientific-abstract-in-six-easy-steps/>

Summarize (in one sentence) why nobody else has adequately answered the research question yet. For a PhD thesis, you'll have an entire chapter, covering the prior work in the literature. Here you have to boil that down to one sentence. But remember, the trick is not to try and cover all the ways in which people have tried and failed; the trick is to explain that there's this one particular approach that nobody else tried yet (hint: it's the thing that your research does). But here you're phrasing it in such a way that it's clear it's a gap in the literature. So use a phrase such as "previous work has failed to address..." (if you're writing a more general essay, you still need to summarize the source material you're drawing on, so you can pull the same trick – explain in one sentence what the general message in the source material is, but expressed in terms of what's missing)

Explain, in one sentence, how you tackled the research question. What's your big new idea? (Again for a more general essay, you might want to adapt this slightly: what's the new perspective you have adopted? or: What's your overall view on the question you introduced in step 2?)

In one sentence, how did you go about doing the research that follows from your big idea. Did you run experiments? Build a piece of software? Carry out case studies? This tends to be the longest sentence, above all in a PhD thesis, which covers years of research. But don't overdo it – we're still looking for a sentence that you could read aloud without having to stop for breath. Remember, the word 'abstract' means a summary of the main ideas with most of the detail left out. So feel free to omit detail! (For those of you who got this far and are still insisting on writing an essay rather than signing up for a PhD, this sentence elaborates on sentence 4 – explore the consequences of your new perspective).

As a single sentence, what's the key impact of your research? Here we're not looking for the outcome of an experiment. We're looking for a summary of the implications. What's it all mean? Why should other people care? What can they do with your research. (Essay folks: all the same questions apply: what conclusions did you draw, and why would anyone care about them?)

DANKSAGUNGEN

Ich möchte mich bei den kleinen grünen Männchen vom Mars bedanken. Ohne ihre Sondiergeräte wären die in dieser Arbeit beschriebenen medizinischen Experimente an den Affen für alle Beteiligten wesentlich unangenehmer gewesen.

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ABKÜRZUNGSVERZEICHNIS

This document is incomplete. The external file associated with the glossary ‘abbreviations’ (which should be called `thesis-example.gls-abr`) hasn’t been created.

Check the contents of the file `thesis-example.gls-abr`. If it’s empty, that means you haven’t indexed any of your entries in this glossary (using commands like `\gls` or `\glsadd`) so this list can’t be generated. If the file isn’t empty, the document build process hasn’t been completed.

Try one of the following:

- Add `automake` to your package option list when you load `glossaries-extra.sty`. For example:

```
\usepackage[automake]{glossaries-extra}
```

- Run the external (Lua) application:

```
makeglossaries-lite.lua "thesis-example"
```

- Run the external (Perl) application:

```
makeglossaries "thesis-example"
```

Then rerun $\LaTeX$ on this document.

This message will be removed once the problem has been fixed.

KAPITEL 1

INTRODUCTION

Purpose and scope of your entire report. The purpose of your entire report is to make a *scientific argument using the scientific method*. A scientific argument always has the following steps that all must come in this order.

SM1 *Explicate the assumptions and state of the art* on which you are going to conduct your research to investigate your research problem/test the hypothesis.

SM2 Clearly and precisely *formulate a research problem or hypothesis*.

SM3 *Describe the (research) method* that you followed to investigate the problem / to test the hypothesis in a way that *allows someone else to reproduce your steps*. The method must include steps and criteria for evaluating whether you answered your question successfully or not.

SM4 *Provide execution details* on how you followed the method in the given, specific situation.

SM5 *Report your results* by describing and summarizing your measurements. You must not interpret your results.

SM6 *Now interpret your results* by contextualizing the measurements and drawing conclusion that lead to answering your research problem or defining further follow-up research problems.

This template marks the parts of the structure with SM1-SM6 to recall to you which step of a scientific argument is used and where.

Purpose and scope of Kapitel 1. The introduction chapter is a summary of your work and your scientific argument that must be understandable to anyone in your scientific field, e.g., anyone in Data Science. A reader must be able to comprehend the problem, method, relevant execution details, results, and their interpretation by reading the introduction and the introduction alone. Section 1.1 introduces the general topic of your research. Section 1.2 discusses the state of the art and identifies a research. Section 1.3 then states the research problem to investigate. Section 1.11 explains the research method that was followed, possibly with execution details. Section 1.5 then presents the results and their interpretation. A reader should need to read further only if they are not convinced or need more details to reproduce your study. The individual chapters and sections provide the details for each of the steps in your scientific argument.

Write the introduction chapter *after* you wrote all other chapters, but you should keep on making notes for each of the sections as you write the later chapters. .

Purpose and scope of the introduction paragraph to a chapter. The paragraph you are reading above is a typical introductory paragraph to a chapter. It is a high-level summary of the chapters' topic (SM1 and SM2). It gives the reader some guidance by breaking down the chapter topic into subtopics that are named (SM3) in the right order with forward references to the corresponding sections (SM4). It may close with announcing the result you obtain (SM6) but the opening paragraph of the introduction is not the place for it.

Zwei Arten von Lesenden. Die Arbeit soll sich von vorne bis hinten lesen lassen: Jeder Begriff wird eingeführt, bevor er verwendet wird. Rückverweise sind in Ordnung; Vorwärtsverweise im Fließtext sollten selten sein. Gleichzeitig sollen Lesende, die die Arbeit irgendwo aufschlagen und zehn Minuten Zeit haben, ihren Weg finden: Jedes Kapitel beginnt mit einem einleitenden Absatz wie oben beschrieben, der die folgenden Abschnitte skizziert, auf das zurückverweist,

worauf das Kapitel aufbaut, und auf die Stellen vorausverweist, an denen seine Ergebnisse verwendet werden. Das Buch „Student Thesis Projects“ [LLDS26] (Open Access) erklärt ausführlich, wie man eine Abschlussarbeit organisiert, durchführt, schreibt und präsentiert. Vor der Abgabe bitte die Checkliste in docs/thesis-checklist.md durchgehen und alle diese Hinweise entfernen.

1.1 Context and Topic (SM1)

Purpose and scope. You begin with providing the general scientific audience an introduction into the specific topic of your work. The aim of this section is to first introduce the *general subject of study* (“Giraffes are well-known animals and everyone’s favorite”), the *specific topic of societal or scientific interest* to investigate (“Giraffes have blue tongues”) and the *objective of society/science towards* this topic (“it is unknown at the moment how the blue color tone evolved”). It must be understandable by the general scientific public. Every *term* with a specific meaning must be highlighted and introduced in precise language/concepts that only builds on a general scientific background.

At the end of this section, you have explained and established a general goal that society/science universally agrees to be worth achieving (“knowing how everyone’s favorite animal evolved the colour of their tongue”).

1.2 State of the Art (SM1)

Purpose and scope. You provide a more in-depth introduction into the research topic by contrasting the current state of the art in society/science in relation to the research topic you introduced in Abschnitt 1.1. This introduction has to

- present established facts, methods, and results that provide a deeper understanding of the research topic (“prior work on giraffe genomes, relevance of giraffes for societal well-being, giraffes being a model-animal for various other studies, etc.”)
- discuss in which ways prior and recent ideas still fall short of reaching the general goal you explained in Abschnitt 1.1 (“prior work only sequenced the genome of one giraffe and did not consider genes of ancient ancestors”)

You have to provide citations/literature references for each of the statements and claims you are making. This section summarizes the related work discussion in Abschnitt 1.6.

At the end of this section, you have established a *knowledge gap* between the state of the art and the general objective you developed in Abschnitt 1.1. *Stating a (knowledge) gap between a status quo and a desired situation is the first step of a writing scientific argument.*

1.3 Research Question (SM2)

Purpose and scope. In this section you state in which way you will address the knowledge gap you identified at the end of Abschnitt 1.2. One work cannot address and resolve the entire knowledge gap. The purpose of this section is to detail the specific part of the knowledge that you will address. You thereby make all the assumptions explicit that underlie your work (“in this report we focus on genomes of female giraffes who lived in the years 1950-2000 in South Africa”).

Your general research question states

- The starting point/assumptions you are making from which your research starts (“for the given 13 genomes of female giraffes...”), and
- the final objective/solution you want to reach (“...identify the genes involved in color expression of giraffe tongues...”)
- and the evaluation criteria that will determine whether you are successful (“...that are present in at least 75% of the studied giraffes”)

Break your general research question down into sub-research questions. You may do this here. The sub-research questions have to form a chain that take you in smaller steps from the starting point/assumptions of your general research question to your final objective and evaluation.

1.4 Method or Approach (SM3, SM4)

Purpose and scope. In this section you outline the method that you applied to answer the research questions, or the new technical approach that you developed to answer it. It is a summary of the steps that someone else has to take in order to reproduce your steps. Mention here the data sets you had to obtain/gather/analyze, interviews with stakeholders you had to make to further develop the research questions, technical artifacts (programs, algorithms, models) you could apply or that you had to develop (and how they work).

The section is most readable if you give each of the steps in your method its own paragraph. In each paragraph you first briefly explain the concept of the step in your method (SM3, “we explored the data through visual analytics”) and then provide details in execution (SM4, “we used tool X, we developed dashboard Y”) include a forward reference to the respective chapter that provides more details.

1.5 Findings (SM5, SM6)

Purpose and scope. You close the introduction by stating the evaluation setup you designed to evaluate the success of your study regarding the research objective, which comes in two steps. It summarizes your evaluation in Abschnitt 1.16.

Results (SM5)

You state the evaluation method that is in line with your research question from Abschnitt 1.3 and summarize the measurements you obtained but you do not interpret them, i.e., you only report the numbers but you do not include judging statements.

Interpretation (SM6)

You summarize your interpretation of the results and draw conclusions. State whether, and to which degree, the research question from Abschnitt 1.3 has been answered.

Close by stating how much closer society and science have come in answering the general objective you outlined in Abschnitt 1.1.

1.6 Background (SM1)

Purpose and scope. The background chapter has multiple roles.

- *Preliminaries.* It has to provide all (and exactly the) information that is necessary to understand the methodological and technical parts of your work in the specific area of study. Assume as starting point another student in your degree who did not study the specific subject you are studying but has the task to understand your work. Which concepts, terms, definitions, etc. does the student have to know? Which formulas, symbols, etc. are standard in this topic? Only introduce definitions if you actually need them in any of the subsequent chapters.
- *Related Work.* It has to provide a comprehensive discussion of all prior work in the area on this subject. Your discussion has to summarize these prior works and has to explain in which way the research question you are solving (Abschnitt 1.3) has not been solved yet because prior work had more limiting assumptions, addressed a different angle, their results are not complete etc. Depending on the subject you are studying, the related work part can be larger and warrant an entire chapter on its own, or be fully concluded within Abschnitt 1.2.

You can close the related work discussion by clarifying the positioning and formulation of your research question (SM2) in relation to all the prior work, making more explicit whether you address an existing research question under different premises or whether you work on a modified or completely new research question.

1.7 Problem Exposition (optional)

Purpose and scope. Introduce the problem context in more detail if Abschnitt 1.1 does not provide all necessary information about the problem to follow the rest of the report. This can include further details on the data you studied, context assumptions and requirements, etc.

If you have to expose the problem in more detail here, then this chapter should also provide a more detailed explanation of research question and the method you are applying, i.e., you can now provide more concrete sub-problems compared to Abschnitt 1.3 more details for the method Abschnitt 1.4 because you now have explained the problem much better. A typical structure can be.

1.8 Context/Business Understanding (SM1)

provide details

1.9 Data Understanding (SM1)

provide details

1.10 Detailed Research Questions (SM2)

provide details based on Abschnitt 1.8 and 1.9

1.11 Detailed Method (SM3)

provide details based on Abschnitt 1.8 and 1.9

1.12 First Real Chapter addressing first Research Problem

Purpose and scope. After you stated research context (SM1), research problem (SM2), and research method (SM3) in Kapitel 1 and possibly Abschnitt 1.7, the remainder of your entire report addresses execution (SM4), results (SM5), and interpretation (SM6). Do this by addressing the sub-problems through scientific arguments that follow the 6 steps SM1-SM6.

Have a short chapter introduction that recalls and explains the first research problem of your thesis. The problem has to show up in the introduction in Abschnitt 1.3 or in Abschnitt 1.10 already. This provides the background (SM1) for this chapter while the first research problem of the thesis becomes the research question/hypothesis (SM2) for this chapter.

Next, explain in the chapter intro how you solve the research problem in this chapter by breaking it down in further sub-problems. By this, you outline the method (SM3) through which you are going to solve the problem of this chapter. This is necessary to give the reader guidance of what's to come in this chapter and how it fits into the thesis as a whole. Explain that you will address the first sub-problem in Abschnitt 1.13 and the second sub-problem in Abschnitt 1.14, etc. The sections then provide the details for execution and results.

1.13 First Sub-Problem

The first paragraph describes the first sub-problem and develops the requirements a solution has to satisfy (SM2 for this section). The requirements have to be based on the knowledge and reasoning developing in the preceding chapters and sections. Try to use an example to illustrate the problem and the desired properties of the solution. Check that every term/concept you use here has already been defined already in a previous section. If you cannot describe your problem without defining new terms, you may have to add another section before this one that develops the terms and concepts you need to explain the problem.

Abbildung 1.1: A scientific figure that has to be explained in the text

The second paragraph describes the method/approach how you address the problem (SM3 for this section). Describe the method in a level of detail that allows another student to reproduce your steps. Make use of appendices if certain details take too much space.

The third, fourth, and following paragraph provides details on applying the method or developing a new approach, i.e., execution (SM4) and may explain results (SM5), i.e. details on the steps needed to reproduce the results.

Results (SM5) come in different forms, e.g., conceptual diagrams, algorithms, tables, charts, a list of articles from a literature research etc. You must reference them (“Abbildung 1.1 shows. ..”) and describe the results in text. If you use diagrams, tables, or charts, you cannot expect the reader to know what to you expect them to see in a diagram, table or chart. Describe to them how to read these, explain the meaning of particular elements, point out special observations. Describe the results, but do not interpret them. Make use of appendices if certain details take too much space.

After describing the results, you may interpret them (SM6). Here you can infer what a particular observation means (for you), how it can be applied, or what others can do with it. Do not interpret your results before you have described all of them. This is a common mistake done by most beginner writers. Get to the point: the final finding or interpretation. But you forget that your reader does not understand yet what you are interpreting - they do not know yet what you do know. Interpret the results only after describing all of them. The interpretation must be based on the written description only. Then you can be sure that your readers can follow your interpretation and reach the same conclusions as you have.

Ideally, your interpretation leads to the next sub-problem in Abschnitt 1.14.

1.14 Second Sub-Problem

You now build on the solution to the first sub-problem of Abschnitt 1.13 (SM1) and recall second sub-problem (SM2, you detailed in the introduction of this chapter) and follow the same pattern as before (SM3-SM6).

Note that not all sections may not include all parts SM1-SM6 in all detail. Some sections need not restate the background (SM1) or the research problem (SM2) if a previous section already defined them. Sometimes, a section is only dedicated to describing the method (SM3) and execution (SM4) and does not contain any results or interpretations. Sometimes results (SM5) and interpretations (SM6) only come in the evaluation chapter.

What is important for you when you are writing a scientific argument is not to slavishly have SM1-SM6 in each section explicitly, but that you are always fully aware of the following:

- Which step of a scientific argument am I currently writing (SM1, SM2, ..., SM6)?
- Does the step that I am writing come in the right order, i.e., if you are writing about execution (SM4, e.g., details of building a model), is there a preceding paragraph or section that describes the method (SM3) and is that one preceded by a clear statement of the (sub-)problem addressed (SM2)?
- Are you really *not* writing interpretation SM6 before SM5, SM4, or SM3?
- Is it clear to the reader which part of the scientific argument you are currently making?

1.15 Second Real Chapter

Have a short chapter introduction that recalls what you already achieved in Abschnitt 1.12 and explain the second research problem of your thesis. The problem has to show up in the introduction in Abschnitt 1.3 or in Abschnitt 1.10 already. etc.

1.16 Evaluation

Purpose and scope. The evaluation chapter should be the most formal and rigorously structured chapter of your thesis as the validity of your evaluation argument depends on it.

1.17 Objective (SM2)

State what you want to evaluate and what you want to measure.

1.18 Setup (SM3)

State which data, participants, tools, etc. you chose and why. State how you measure outcomes and how you compare them to baselines, reference groups, etc.

1.19 Execution (SM4)

Provide all details on the execution that are necessary to allow another person to reproduce your results at a later point.

1.20 Results (SM5)

You only report the measurements. You must present and reference them (“Abbildung 1.2 shows. ..”) and describe the results in text. If you use diagrams, tables, or charts, you cannot expect the reader to know what to expect them to see in a diagram, table or chart. Describe to them how to read these, explain the meaning of particular elements, point out special observations. Describe the results, but do not interpret them. Make use of appendices if certain details take too much space.

Abbildung 1.2: Another scientific figure that has to be explained in the text

1.21 Discussion (SM6)

Interpret the results only after describing all of them. The interpretation must be based on the written description in Abschnitt 1.20 only. Then you can be sure that your readers can follow your interpretation and reach the same conclusions as you have.



L^AT_EX HINWEISE

Hier sollen allgemeine $\text{\LaTeX}$ -Hinweise gegeben werden, damit man Minimalbeispiele vorliegen hat, um sofort loszulegen.

2.1 Trennung von Absätzen

Pro Satz eine neue Zeile. Das ist wichtig, um sauber versionieren zu können. In $\text{\LaTeX}$ werden Absätze durch eine Leerzeile getrennt. Analogie zu Word: Bei Word werden neue Absätze durch einmal Eingabetaste herbeigeführt. Dies führt bei $\text{\LaTeX}$ jedoch nicht zu einem neuen Absatz, da $\text{\LaTeX}$ direkt aufeinanderfolgende Zeilen zu einer Zeile zusammenfügt. Mächte man nun einen Absatz haben, muss man zweimal die Eingabetaste drücken. Dies führt zu einer leeren Zeile. In Word gibt es die Funktion Großschreibetaste und Eingabetaste gleichzeitig. Wenn man dies drückt, wird einer harter Umbruch erzwungen. Der Text fängt am Anfang der neuen Zeile an. In $\text{\LaTeX}$ erreicht man dies durch Doppelbackslashes (`\`) erzeugt.

Dies verwendet man quasi nie.

Folglich werden neue Absätze insbesondere *nicht* durch Doppelbackslashes erzeugt. Beispielsweise begann der letzte Satz in einem neuen Absatz. Andrew Stacey begründet dies ausführlich im Abschnitt „Line Wrapping“ seines *Document Revision System*².

²<https://loopspace.mathforge.org/HowDidIDoThat/TeX/VCS/>

Zugehöriger L^AT_EX-Quelltext aus ./thesis-example.tex

æEúÚóŒ' , _ , , , , , , , , " , , , _ , , , , _ , ' , _ , , , ,

```
1686 Pro Satz eine neue Zeile.
```

Das ist wichtig, um sauber versionieren zu können.

1688 In LaTeX werden Absätze durch eine Leerzeile getrennt.

Analogie zu Word: Bei Word werden neue Absätze durch einmal

↪ Eingabetaste herbeigeführt.

1690 Dies führt bei LaTeX jedoch nicht zu einem neuen Absatz, da LaTeX

↪ direkt aufeinanderfolgende Zeilen zu einer Zeile

↪ zusammenfügt.

Möchte man nun einen Absatz haben, muss man zweimal die

↪ Eingabetaste drücken.

```
1692 Dies führt zu einer leeren Zeile.
```

In Word gibt es die Funktion Großschreibetaste und Eingabetaste

↪ gleichzeitig.

1694 Wenn man dies drückt, wird einer harter Umbruch erzwungen.

Der Text fängt am Anfang der neuen Zeile an.

1696 In LaTeX erreicht man dies durch Doppelbackslashes (\

↪ `textbackslashtextbackslash`) erzeugt.

//

1698 Dies verwendet man quasi nie.

1700 Folglich werden neue Absätze insbesondere **\emph{nicht}** durch

↪ Doppelbackslashes erzeugt.

Beispielsweise begann der letzte Satz in einem neuen Absatz.

1702 Andrew Stacey begründet dies ausführlich im Abschnitt \enquote{

↪ Line Wrapping} seines \href{https://loopspace.mathforge.}

↪ `org/HowDidIDoThat/TeX/VCS/#section.3`}{\emph{Document

↪ Revision System}}\footnote{\url{https://loopspace.}}

↪ [`mathforge.org/HowDidIDoThat/TeX/VCS/`](http://mathforge.org/HowDidIDoThat/TeX/VCS/)}}

2.2 Notes separated from the text

The package `mindflow` enables writing down notes and annotations in a way so that they are separated from the main text.

Zugehöriger $\text{\LaTeX}$ -Quelltext aus ./thesis-example.tex

```
1710 \begin{mindflow}
1712 This is a small note.
      \end{mindflow}
```

2.3 Umgang mit TODOs

Markierter Text.

Zugehöriger $\text{\LaTeX}$ -Quelltext aus ./thesis-example.tex

```
1718 \textmarker{Markierter Text.}
```

Bei `\textmarker` wird nur die Textfarbe geändert, da dies auch bei einigen Worten gut funktioniert.

Markierter Text.

Zugehöriger $\text{\LaTeX}$ -Quelltext aus ./thesis-example.tex

```
1724 \textcomment{Markierter Text.}{Kommentar dazu.}
```

In Gelb hervorgehoben. Provided indirectly by pdfcomment.sty (soulpos).

Zugehöriger $\LaTeX$ -Quelltext aus ./thesis-example.tex

```
1728 \hl{In Gelb hervorgehoben.}
1730 Provided indirectly by pdfcomment.sty (soulpos).
```

Manuelle Markierung für Text, der seit der letzten Version geändert wurde.

Zugehöriger $\LaTeX$ -Quelltext aus ./thesis-example.tex

```
1734 \modified{Manuelle Markierung für Text, der seit der letzten
    ↪ Version geändert wurde.}
```



Das ist ein Text. **Geänderter Text.**

Zugehöriger $\LaTeX$ -Quelltext aus ./thesis-example.tex

```
1738 Das ist ein Text.
    \change{FL1: Text angepasst}{Geänderter Text}.
```



Hier nur ein Kommentar.

Zugehöriger $\LaTeX$ -Quelltext aus ./thesis-example.tex

```
1742 Hier nur ein Kommentar\sidecomment{Kommentar}.
```



TODO!

Zugehöriger $\LaTeX$ -Quelltext aus ./thesis-example.tex

```
1746 \todo{Hier muss noch kräftig Text produziert werden}
```


Zugehöriger $\LaTeX$ -Quelltext aus ./thesis-example.tex

```

1768  Numbers can be written plain text (such as 100), by using the \
    ↪ href{https://ctan.org/pkg/siunitx}{siunitx} package as
    ↪ follows:
1770  \SI{100}{\km\per\hour},
    or by using plain \LaTeX{} (and math mode):
1772  $100 \frac{\mathit{km}}{h}$.

```

5 % of 10 kg

Zugehöriger L^AT_EX-Quelltext aus ./thesis-example.tex

1776 \SI{5}{\percent} of \SI{10}{kg}

Numbers are automatically grouped: 123 456.

Zugehöriger L^AT_EX-Quelltext aus ./thesis-example.tex

```

1780 Numbers are automatically grouped: \num{123456}.

```

2.6 Surrounding Text by Quotes

Please use the „enquote command“ to quote something. Quoting with „quote“ or “quote” also works.

Zugehöriger L^AT_EX-Quelltext aus ./thesis-example.tex

1786 Please use the `\enquote{enquote command}` to quote something.
Quoting with `"`quote"` or ``quote'` also works.

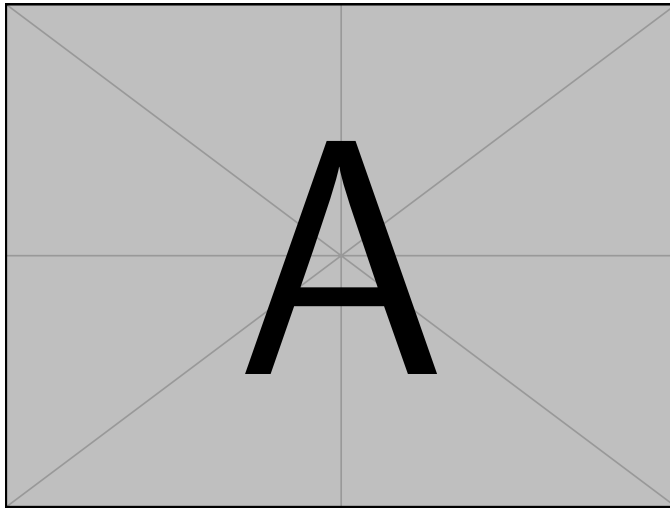


Abbildung 2.1: Example figure for zceref demo

Heading1	Heading2
One	Two
Three	Four

Tabelle 2.1: Example table for zceref demo

2.7 Zref-clever examples

zref-clever demonstration: Zceref (short for zceref[S]) at beginning of sentence, zceref in all other cases.

Abbildung 2.1 shows a simple fact, although Abbildung 2.1 could also show something else.

Tabelle 2.1 shows a simple fact, although Tabelle 2.1 could also show something else.

Abschnitt 2.7 shows a simple fact, although Abschnitt 2.7 could also show something else.

Zugehöriger $\text{\LaTeX}$ -Quelltext aus `./thesis-example.tex`

```

1818 \Zcref{fig:ex:cref} shows a simple fact, although \zcref{fig:ex:
      ↪ cref} could also show something else.

1820 \Zcref{tab:ex:cref} shows a simple fact, although \zcref{tab:ex:
      ↪ cref} could also show something else.

1822 \Zcref{sec:ex:cref} shows a simple fact, although \zcref{sec:ex:
      ↪ cref} could also show something else.

```

2.8 Formulas and Equations

Equations $f(x) = x$ inside the text can be provided.

Zugehöriger $\text{\LaTeX}$ -Quelltext aus `./thesis-example.tex`

```

1828 Equations  $f(x)=x$  inside the text can be provided.

```

A list with all available mathematical symbols is provided at <http://texdoc.net/pkg/symbols-a4>.

As an example, the set of natural numbers is given by $\mathbb{N}$.

Zugehöriger $\text{\LaTeX}$ -Quelltext aus `./thesis-example.tex`

```

1834 As an example, the set of natural numbers is given by  $\mathbb{N}$ 
      ↪  $\mathbb{N}$ .

```

For the documentation of editing mathematical formulas, read the package documentation of `amsmath`³.

³<http://texdoc.net/pkg/amsmath>

Gleichung (2.1) is numbered and can be referenced in the text:

$$x = y \tag{2.1}$$

Zugehöriger $\LaTeX$ -Quelltext aus ./thesis-example.tex

```
1840 \begin{align}
1842   \label{eq:test}
      x = y
1844 \end{align}
```

The following equation is not numbered because of using `\align*` as environment.

$$x = y$$

Zugehöriger $\LaTeX$ -Quelltext aus ./thesis-example.tex

```
1848 \begin{align*}
1850   x = y
      \end{align*}
```

The template offers `\abs` to enable the bars to scale well at the absolute value: $|X|$.

Zugehöriger $\LaTeX$ -Quelltext aus ./thesis-example.tex

```
1856 $\abs{X}$.
```

The documentation available at <http://www.ctan.org/tex-archive/help/Catalogue/entries/voss-mathmode.html> provides more details about mathematical environments.

Füge deine Abbildung hier ein.

Abbildung 2.2: Bildunterschrift.

2.9 Abbildungen

Abbildung 2.2 zeigt etwas Interessantes

Zugehöriger $\text{\LaTeX}$ -Quelltext aus ./thesis-example.tex

```
1864 \Zcref{fig:label} zeigt etwas Interessantes

1866 \begin{figure}
      \centering
1868   Füge deine Abbildung hier ein.
      \caption{Bildunterschrift.}
1870   \label{fig:label}
      \end{figure}
```

Fotos im JPEG- oder PNG-Format werden genauso eingebunden, wie Abbildung 2.3 zeigt.

Zugehöriger $\text{\LaTeX}$ -Quelltext aus ./thesis-example.tex

```
1876 \Zcref{fig:label} zeigt etwas Interessantes

1878 \begin{figure}
      \centering
      \includegraphics[width=.5\linewidth]{figures/ulm-minster.jpg}
1880   \caption{Ulmer Münster. Foto: wkrauth, CC0, via Wikimedia
            ↪ Commons.}
      \label{fig:jpg}
1882 \end{figure}
```



Abbildung 2.3: Ulmer Münster. Foto: wkrauth, CC0, via Wikimedia Commons.

2.10 Sub Figures

An example of two sub figures is shown in Abbildung 2.4.

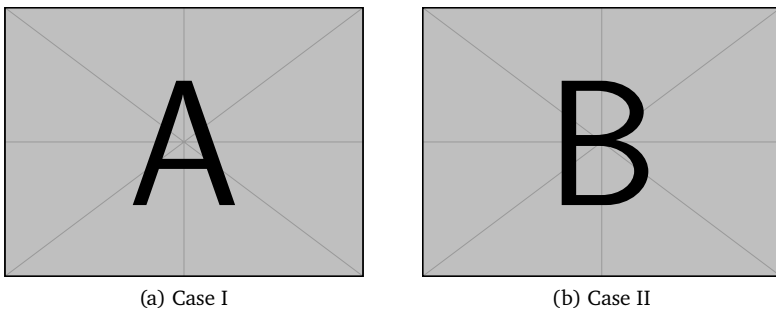


Abbildung 2.4: Example figure with two sub figures.

```

1890 \begin{figure}[!b]
      \centering
1892 \subfloat[Case I]{\includegraphics[width=.4\linewidth]{example-
      ↪ image-a}%
      \label{fig:first_case}}
1894 \hfil
      \subfloat[Case II]{\includegraphics[width=.4\linewidth]{example
      ↪ -image-b}%
1896 \label{fig:second_case}}
      \caption{Example figure with two sub figures.}
1898 \label{fig:two_sub_figures}
      \end{figure}

```

2.11 Figures with tikz

The tikz package creates graphics programmatically. This package generates grids and other regular structures.

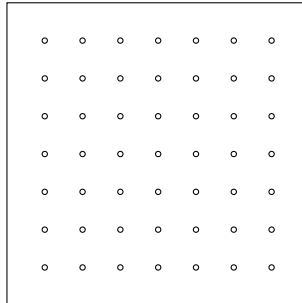


Abbildung 2.5: A regular grid generated easily with two for loops.

Zugehöriger $\text{\LaTeX}$ -Quelltext aus ./thesis-example.tex

```

1908 \begin{figure}[ht]
      \centering
1910 \begin{tikzpicture}
      \draw(0,0) rectangle (4,4);
1912 \foreach \x in {0.5,1,1.5,2,2.5,3,3.5}
      \foreach \y in {0.5,1,1.5,2,2.5,3,3.5}
1914 \draw(\x,\y) circle (1pt);
      \end{tikzpicture}
1916 \caption{A regular grid generated easily with two for loops.}\
      \label{fig:tikz_example}
\end{figure}

```

2.12 Plots with pgfplots

The pgfplots package plots functions directly in $\text{\LaTeX}$, similar to MATLAB or gnuplot. Some visual examples are available in the package documentation⁴.

Zugehöriger $\text{\LaTeX}$ -Quelltext aus ./thesis-example.tex

```

1926 \begin{figure}[ht]
      \centering
1928 \begin{tikzpicture}
      \begin{axis}[xlabel=$x$, ylabel=$\sin(x)$]
1930 \addplot {sin(deg(x))}; % Plot the sine function
      \end{axis}
1932 \end{tikzpicture}
      \caption{Plot of  $\sin(x)$  directly inside the figure
      \hookrightarrow environment with pgfplots.}
1934 \label{fig:pgfplots_example}
\end{figure}

```

⁴<http://texdoc.net/pkg/pgfplots>

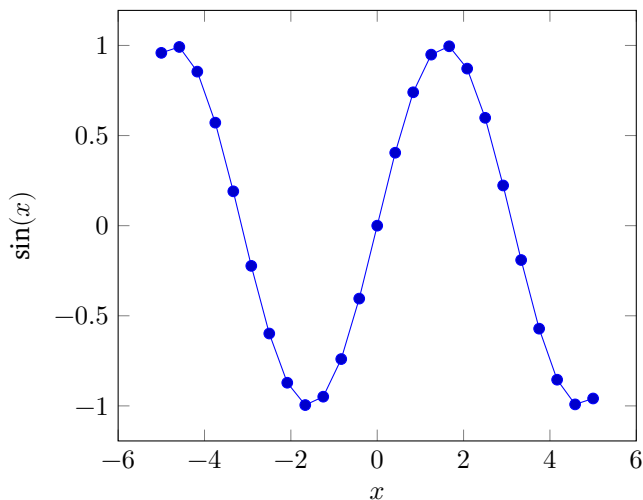


Abbildung 2.6: Plot of $\sin(x)$ directly inside the figure environment with pgfplots.

Coordinates can also be given explicitly instead of as a function:

Zugehöriger $\text{\LaTeX}$ -Quelltext aus ./thesis-example.tex

```

1940 \begin{figure}[ht]
      \centering
1942 \begin{tikzpicture}
      \begin{axis}[xlabel=$x$, ylabel=$y$]
1944 \addplot coordinates {(0,0) (1,1) (2,4) (3,9)}; % Plot
           ↪ explicit coordinates
      \end{axis}
1946 \end{tikzpicture}
      \caption{Explicit coordinates plotted with pgfplots.}
1948 \label{fig:pgfplots_coords}
      \end{figure}

```

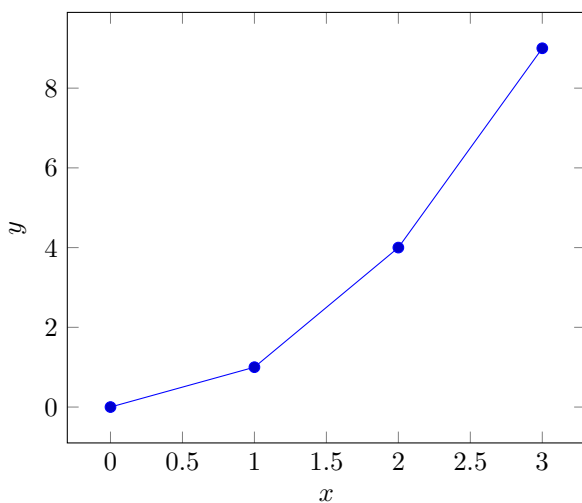


Abbildung 2.7: Explicit coordinates plotted with pgfplots.

Tabelle 2.2: Simple Table

Heading1	Heading2
One	Two
Three	Four

Tabelle 2.3: Table with diagonal line

Row \ Column	A	B
	x	y

2.13 Tables

Zugehöriger $\text{\LaTeX}$ -Quelltext aus ./thesis-example.tex

```
1954 \begin{table}
1955 \caption{Simple Table}
1956 \label{tab:simple}
1957 \centering
1958 \begin{tabular}{ll}
1959 \toprule
1960 Heading1 & Heading2 \\
1961 \midrule
1962 One & Two \\
1963 Three & Four \\
1964 \bottomrule
1965 \end{tabular}
1966 \end{table}
```

Zugehöriger $\text{\LaTeX}$ -Quelltext aus ./thesis-example.tex

```
1970 % Adapted from https://tex.stackexchange.com/a/468994/9075
1972 \begin{table}
      \caption{Table with diagonal line}
1974 \label{tab:diag}
      \begin{center}
1976 \begin{tabular}{|l|c|c|}
          \hline
1978 \diagbox{Row}{Column} & A & B \\
          \hline
1980 1 & x & y \\
          \hline
1982 \end{tabular}
      \end{center}
1984 \end{table}
```

2.14 Tables spanning multiple pages

The longtable package typesets tables that automatically break across page boundaries. The header (\endhead) and footer (\endfoot) rows are repeated on every page; once enough rows are added, the table spans more than one page on its own.

Tabelle 2.4: A sample long table.

First column	Second column	Third column
A	BC	D
A	BC	D
A	BC	D
A	BC	D
A	BC	D
Continued on next page		

Tabelle 2.4 – continued from previous page

First column	Second column	Third column
A	BC	D
A	BC	D
A	BC	D
A	BC	D
A	BC	D
A	BC	D
A	BC	D

```

1994 \begin{longtable}{|l|l|l|l|}
      \caption{A sample long table.}\label{tab:long} \\
1996 \hline
      \multicolumn{1}{|c|}{\textbf{First column}} & \multicolumn{1}{c}{c}
          \hookrightarrow |}{\textbf{Second column}} & \multicolumn{1}{c|}{\
          \hookrightarrow \textbf{Third column}} \\ \hline
1998 \endfirsthead

2000 \multicolumn{3}{c}{\bfseries \tablename\ \thetable{} --
          \hookrightarrow continued from previous page}} \\
      \hline
2002 \multicolumn{1}{|c|}{\textbf{First column}} & \multicolumn{1}{c}{c}
          \hookrightarrow |}{\textbf{Second column}} & \multicolumn{1}{c|}{\
          \hookrightarrow \textbf{Third column}} \\ \hline
      \endhead

2004 \hline \multicolumn{3}{|r|}{\text{Continued on next page}} \\ \hline
2006 \endfoot

2008 \hline\hline
      \endlastfoot

2010 A & BC & D \\
2012 A & BC & D \\
      A & BC & D \\
2014 A & BC & D \\
      A & BC & D \\
2016 A & BC & D \\
      A & BC & D \\
2018 A & BC & D \\
      A & BC & D \\
2020 A & BC & D \\
      A & BC & D \\
2022 A & BC & D \\
      \end{longtable}

```

2.15 Quellcode

Listing 2.1 zeigt XML-Quelltext. Zeile 2 enthält einen Kommentar.

```
<listing name="example">
2  <!-- comment -->
    <content>not interesting</content>
4 </listing>
```

Listing 2.1: Beispiel-XML-Listing

Zugehöriger L^AT_EX-Quelltext aus ./thesis-example.tex

```

2028 \Zcref{lst:XML} zeigt XML-Quelltext.
2030 \Zcref{line:comment} enthält einen Kommentar.

2032 \begin{lstlisting}[
    language=XML,
2034    caption={Beispiel-XML-Listing},
    label={lst:XML}]
2036 <listing name="example">
    <!-- comment --> (* \label{line:comment} *)
2038    <content>not interesting</content>
    </listing>
2040 \end{lstlisting}

```

Der zusätzliche Parameter `float` führt dazu, dass das Listing auch `floatet`. Listing 2.2 zeigt das gleitende Listing.

```
1 <listing name="example">
    Floating
3 </listing>
```

Listing 2.2: Beispiel-XML-Listing – gleitend

```
1 {
2   key: "value"
3 }
```

Listing 2.3: Beispiel-JSON-listing

Zugehöriger L^AT_EX-Quelltext aus ./thesis-example.tex

```
2046 \begin{lstlisting}[
2048   % Es ist möglicih, die Abstände bei Bedarf einzustellen
2050   % aboveskip=2.5\baselineskip,
2052   % belowskip=-.8\baselineskip,
2054   float,
2056   language=XML,
2058   caption={Beispiel-XML-Listing -- gleitend},
2059   label={lst:flXML}]
2060 <listing name="example">
2061   Floating
2062 </listing>
2063 \end{lstlisting}
```

Es ist möglich auch JSON zu setzen, wie in Listing 2.3 gezeigt.

```

1 public class Hello {
    public static void main (String[] args) {
3         System.out.println("Hello World!");
    }
5 }

```

Listing 2.4: Example Java listing

Zugehöriger L^AT_EX-Quelltext aus ./thesis-example.tex

```

2064 \begin{lstlisting}[
    float,
2066     language=json,
    caption={Beispiel-JSON-listing},
2068     label={lst:json}]
    {
2070     key: "value"
    }
2072 \end{lstlisting}

```

Java ist auch möglich – Listing 2.4.

```

2078 \begin{lstlisting}[
        caption={Example Java listing},
2080     label=lst:java,
        language=Java,
2082     float]
    public class Hello {
2084         public static void main (String[] args) {
            System.out.println("Hello World!");
2086         }
    }
2088 \end{lstlisting}

```

Listings longer than the remaining space continue on the next page if they are *not* floating. A floating listing is always placed on a single page. Listing 2.5 spans multiple pages.

```

1  public class Squares {

3      /** Returns the square of 1. */
      public int square1() {
5          return 1 * 1;
      }

7      /** Returns the square of 2. */
9      public int square2() {
      return 2 * 2;
11     }

13     /** Returns the square of 3. */
      public int square3() {
15         return 3 * 3;
      }

17     /** Returns the square of 4. */

```

```

19     public int square4() {
20         return 4 * 4;
21     }

22
23     /** Returns the square of 5. */
24     public int square5() {
25         return 5 * 5;
26     }

27
28     /** Returns the square of 6. */
29     public int square6() {
30         return 6 * 6;
31     }

32
33     /** Returns the square of 7. */
34     public int square7() {
35         return 7 * 7;
36     }

37
38     /** Returns the square of 8. */
39     public int square8() {
40         return 8 * 8;
41     }

42
43     /** Returns the square of 9. */
44     public int square9() {
45         return 9 * 9;
46     }

47
48     /** Returns the square of 10. */
49     public int square10() {
50         return 10 * 10;
51     }

52
53     /** Returns the square of 11. */
54     public int square11() {
55         return 11 * 11;

```

```

    }
57
    /** Returns the square of 12. */
59    public int square12() {
        return 12 * 12;
61    }

63    public static void main(String[] args) {
        System.out.println(new Squares().square12());
65    }
}

```

Listing 2.5: Long Java listing spanning multiple pages

2.16 Itemization

One can list items as follows:

- Item One
- Item Two

Zugehöriger $\LaTeX$ -Quelltext aus `./thesis-example.tex`

```

2172 \begin{itemize}
        \item Item One
2174 \item Item Two
    \end{itemize}

```

With the package `paralist`, one can create itemizations with lesser spacing:

- Item One
- Item Two

1. All these items... 2. ...appear in one line 3. This is enabled by the paralist package.

Zugehöriger $\text{\TeX}$ -Quelltext aus `./thesis-example.tex`

```

2208 \begin{inparaenum}
      \item All these items...
2210 \item ...appear in one line
      \item This is enabled by the paralist package.
2212 \end{inparaenum}

```

2.17 Abkürzungen

Mit `\gls{...}` können Abkürzungen eingebaut werden, beim ersten Aufrufen wird die lange Form eingesetzt. Beim wiederholten Verwenden von `\gls{...}` wird automatisch die Kurzform angezeigt. Außerdem wird die Abkürzung automatisch in die Abkürzungsliste eingefügt. Mit `\glspl{...}` wird die Pluralform verwendet. Möchte man, dass bei der ersten Verwendung direkt die Kurzform erscheint, so kann man mit `\glsunset{...}` eine Abkürzung als bereits verwendet markieren. Das Gegenteil erreicht man mit `\glsreset{...}`.

Definiert werden Abkürzungen in der Datei *abbreviationstex* mithilfe von `\newacronym{...}{...}{...}`.

Mehr Infos unter: <https://ctan.org/pkg/bib2gls>.

Beim ersten Durchlauf betrug die Fehlerrate (FR) 5. Beim zweiten Durchlauf war die FR 3. Die Pluralform sieht man hier: FRs. Um zu demonstrieren, wie das Abkürzungsverzeichnis bei längeren Beschreibungstexten aussieht, muss hier noch RDBMS (RDBMS) erwähnt werden.

Deutschsprachige Anwendervereinigung $\text{\TeX}$ e.V. (DANTE e.V.) is a local $\text{\TeX}$ user group. The German-speaking local $\text{\TeX}$ user group is DANTE e.V.. A General Practitioner (G.P.) is a medical doctor. I went to my surgery to see the G.P..

Zugehöriger L^AT_EX-Quelltext aus ./thesis-example.tex

2228 Beim ersten Durchlauf betrug die `\gls{fr}` 5.

2230 Beim zweiten Durchlauf war die `\gls{fr}` 3.

Die Pluralform sieht man hier: `\glspl{fr}`.

2232 Um zu demonstrieren, wie das Abkürzungsverzeichnis bei längeren
 ↪ Beschreibungstexten aussieht, muss hier noch `\glspl{rdbms}`
 ↪ } erwähnt werden.

2234 `\gls{dante}` is a local `\TeX` user group.

The German-speaking local `\TeX` user group is `\gls{dante}`.

2236 A `\gls{gp}` is a medical doctor.

I went to my surgery to see the `\gls{gp}`.

2.18 Other Features

The words „workflow“ and „dwarflike“ can be copied from the PDF and pasted to a text file.

Zugehöriger L^AT_EX-Quelltext aus ./thesis-example.tex

2242 The words \enquote{workflow} and \enquote{dwarflike} can be
 ↳ copied from the PDF and pasted to a text file.

The symbol for powerset is now correct: $\wp$ and not a Weierstrass p ($\wp$).

2.19 Zref-vario examples

zref-vario demonstration: `zvref[S]` at beginning of sentence, `zvref` in all other cases.

Abbildung 2.1 auf Seite 42 shows a simple fact, although Abbildung 2.1 auf Seite 42 could also show something else.

Tabelle 2.1 auf Seite 42 shows a simple fact, although Tabelle 2.1 auf Seite 42 could also show something else.

Abschnitt 2.7 auf Seite 42 shows a simple fact, although Abschnitt 2.7 auf Seite 42 could also show something else.

Zugehöriger $\text{\LaTeX}$ -Quelltext aus `./thesis-example.tex`

```
2272 \zvref[S]{fig:ex:cref} shows a simple fact, although \zvref{fig:
      ↪ ex:cref} could also show something else.

2274 \zvref[S]{tab:ex:cref} shows a simple fact, although \zvref{tab:
      ↪ ex:cref} could also show something else.

2276 \zvref[S]{sec:ex:cref} shows a simple fact, although \zvref{sec:
      ↪ ex:cref} could also show something else.
```

2.20 Citations

When referencing something from the bibliography file, it will automatically appear in the references section. If a reference is not cited, it is not appearing there.

Standard case: Citing indirectly citing something [Sch17]. In case one wants to name the author: Scharrer [Sch17] shows a minimal $\text{\LaTeX}$ example.

Zugehöriger $\LaTeX$ -Quelltext aus ./thesis-example.tex

```
2284 \textsuperscript{st}“ „nd“, ...: E.g., „The 2nd  
Standard case: Citing indirectly citing something~\cite{mwe}.  
2286 In case one wants to name the author: \textcite{mwe} shows a  
↪ minimal \LaTeX{} example.
```

Note that `\textcite{mwe}` prints both the author and the reference to the bibliography entry.

Remember that you have to call `biber thesis-example` to generate the bibliography data for `lualatex`. You will need to run `lualatex` twice so that the page numbers are up to date.

In the bibliography, use `\textsuperscript` for „st“, „nd“, ...: E.g., „The 2nd conference on examples“. When you use `JabRef`, you can use the `clean up` command to achieve that. See <https://help.jabref.org/en/CleanupEntries> for an overview of the `cleanup` functionality.

2.21 Miscellaneous Examples

Reference test: Abschnitt 2.21, Abbildung 2.8 and Algorithmus 2.1.

Checkmark: ✓. Crossmark: ✗.

Zugehöriger $\LaTeX$ -Quelltext aus ./thesis-example.tex

```
\dingcheck.  
2306 Checkmark: \dingcheck.  
Crossmark: \dingcross.
```

... missing figure ...

Abbildung 2.8: Figure

... missing figure ...

Abbildung 2.9: Rotated figure

2.21.1 Algorithms

Algorithmus 2.1 *algo*

1: $a \leftarrow 0$
2: State 2

Algorithmus 2.2 Algorithm 2

1: $a \leftarrow 0$
2: State 2

Algorithmus 2.1 has already shown an algorithm. Test of line referencing:
line 2 (Algorithmus 2.1) and line 2 (Algorithmus 2.2).

2.21.2 Definitions

Definition 2.1 (Title)

Definition Text

Definition 2.1 shows ...

2.21.3 Enumerations

- a) a
- b) b
- c) c
- d) d

Equivalent to paralist's inparaenum: a) a b) b c) c d) d

first First

second Second

third Third

first First

second Second

third Third

first First

second Second

third Third

**first label with a long description text breaking over one line. Enabled by
enumitem package** First

second Second

third Third

**first label with a long description text breaking over one line. Defined in
template.tex**

First

second

Second

third

Third

- First
- Second
- Third

The optional parameter changes the marker that is prepended. See <http://www.weinelt.de/latex/item.html>.

A First

B Second

C Third

Incorrect use of the optional parameter as follows:

first First

second Second

third Third

Note that, when `enumitem` is loaded, this behaves differently than with `paralist`.

2.21.4 `fquote`

“ *The program will be finished by next Friday.* ”
— T. SCIENTIST

” *The program will be finished by next Friday.* “
—T. SCIENTIST

KAPITEL 3

ZUSAMMENFASSUNG UND AUSBLICK

Your conclusions are more than a factual summary of your work: they position, interpret, and defend your findings against the state of the art that you discussed in Abschnitt 1.2. You specifically outline which concrete findings or methodological contributions advance our knowledge towards the general objective you introduced in Abschnitt 1.1. Objectively discuss which parts you solved and in which parts you failed.

You should explicitly discuss limitations and shortcomings of your work and detail what kind of future studies are needed to overcome these limitations. Be specific in the sense that your arguments for future work should be based on concrete findings and insights you obtained in your report.

LITERATUR

- [LLDS26] H. Lichter, J. Ludewig, M. Deininger, K. Schneider. *Student Thesis Projects: A Guide to Organizing, Conducting, and Presenting Student Thesis Projects Using the Example of Computer Science*. 7. Aufl. Zollikon: vdf Hochschulverlag, 2026. ISBN: 978-3-7281-4241-2. DOI: 10.3218/4241-2. URL: <https://vdf.ch/product/student-thesis-projects-en.html> (besucht am 23.09.2026) (siehe S. 25).
- [Sch17] M. Scharrer. *The mwe Package*. 2017. URL: <http://texdoc.net/mwe> (siehe S. 65).

Alle Links wurden zuletzt am 29.03.2021 geprüft.



MY FIRST APPENDIX

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NUTZUNG VON KI-TOOLS

KI-Tool	Genutzt für	Genutzt in
ChatGPT5	Erstellung von Abbildungen	Abbildungen 2.2 und 2.3
Google AI search	Pseudocode für Algorithmus XYZ	Algorithmus 2.1
DeepL Write	Umformulierung der Einleitung	Kapitel 1